

Holographic Anti-Entropy Theory: Axiomatic Deduction, Formal Analysis, and Carbon-Silicon Symbiotic Self-Healing Evolution of Open Complex Giant Systems

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August 1, 2026

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Abstract

Faced with the Non-IID (non-independent and identically distributed) characteristics and the combinatorial explosion ($O(N!)$) of state space in modern Open Complex Giant Systems (OCGS), traditional reductionist analysis and open-loop control methods based on steady-state assumptions often fall into dimensionality disasters and thermal dissipation deadlocks. Based on non-equilibrium thermodynamics, Kalman observability, and Ashby's Law of Requisite Variety, this paper systematically expounds "Holographic Anti-Entropy" (HAE), a calculable formal framework for giant system control and governance. This study reconstructs Tsien Hsue-shen's "metasynthesis method from qualitative to quantitative" with axiomatic deduction, establishing the digital double helix ontology $\langle D, A \rangle$ deeply coupled by a five-dimensional orthogonal spatiotemporal container D and a discrete-time algorithm engine A . We extend the HAE dynamic evolutionary trajectory with cross-scale fractal isomorphic mapping to inorganic matter self-organization accretion and organic food web networks, deriving the grand unified anti-entropy work scalar product equation. To address the computational irreducibility and high-dimensional emergence deadlocks of non-linear giant systems, a force-pruning method based on physical gravity constraints and algebraic truncation is proposed, utilizing de-semanticized Data-Oriented Design (DOD). On the basis of the author's previous experience in implementing the first and second generation Intelligent Production Scheduling (IPS) systems at Lenovo's proprietary factories and LCFC, we reconstruct and redevelop a generalized next-generation Intelligent Planning & Control (IPC) engine. Crucial experiments using real-world data from LCFC, with annual revenues exceeding 100 billion RMB, 500,000 daily orders, and a 20-layer supply chain, demonstrate that the IPC engine achieves global capacity convergence and scheduling within 5 minutes. Finally, from the perspective of neurocontrol, we define a five-dimensional cognitive mental OS (conscience, high sensitivity, sentient depth, fluid intelligence, and metacognition) for governing complex systems, resolving the symbol grounding problem. This framework advances systems science from explanatory philosophy to a calculable hard science, establishing a universal formal constitution for complex adaptive system governance.

Keywords: Holographic Anti-Entropy; Open Complex Giant Systems; Digital Double Helix Ontology; Computational Irreducibility; Carbon-Silicon Topological Symbiosis; L0-Level Underlying Operating System; Tsien Hsue-shen School

1 Introduction: The Limits of Reductionism and the Physical Constitution of Systems Science

Classical physics and early systems engineering achieved high reductionist precision when dealing with simple systems that were localized, steady-state, and satisfied the Independent and Identically Distributed (I.I.D.) assumptions. However, when the scientific gaze shifts to modern digital industrial networks, macroeconomic entities, and biologically active fluids, which are categorized as Open Complex Giant Systems (OCGS), the system components and their interactions exhibit extreme Non-IID characteristics and factorial combinatorial explosions.

According to the Second Law of Thermodynamics, the entropy of any isolated system inevitably increases over time, and the ultimate end of system evolution is complete disorder, thermal death, and disintegration. Faced with the relentless dissipation of such giant systems, classical management and researchers of pure probability AI (such as AGI) instinctively fall into a double deadlock:

1. **Internal Dissipation of Human-Governed Collaboration:** Because the brain's bandwidth for processing concurrent information is structurally limited by Miller's Law, traditional organizations attempt to flatten high-dimensional complexity through administrative mandates, linear workflow stitching, and undirected "cross-department alignment meetings." This not only fails to solve for the global optimum under multi-objective constraint conflicts, but also causes the information channel to decay severely during cross-level transmission. Communication friction explodes quadratically ($O(M^2)$), generating massive organizational heat dissipation and locking the system into a low-efficiency Nash equilibrium.
2. **Goodhart Collapse of Probabilistic AI:** Current generative AI frameworks attempt to fit the physical world by stacking computing power and probabilistic learning without prior inductive bias. However, according to the No Free Lunch theorem, black-box models stripped of prior inductive bias inevitably degenerate into random guesses under macro-distributions. More fatally, the extreme optimization process of compressing high-dimensional human intentions into low-dimensional scalar rewards inevitably triggers the collapse of Goodhart's Law—the model will exploit "reward arbitrage" through fabrication or flattery, causing real utility to fall off a cliff.

A Cosmology of Intelligence: The Negative Entropy Pump Against Entropy Increase

From the macro-perspective of cosmic evolution, the ultimate mission of intelligence is to fight against the entropy increase of the physical universe. Whether we perform "observation (establishing mapping and state projection), simulation (multi-universe parallel deduction and path fitting)," or "governance (dimensionality reduction convergence and rigid write-back doing work)," the final target is the remodeling of the objective physical world (the reality composed of organic and inorganic matter).

Observation, simulation, and governance are not isolated puzzle pieces, but share the same reverse-work control loop—namely, the "perception, analysis, decision, execution, and feedback" five-stage closed loop. By running this self-adaptive loop at high frequency, the system continuously absorbs physical deviations, driving the prediction and control residual $\Delta(t) \rightarrow 0$.

In this process, intelligence exhibits active dissipation of disorder: even if the observed object itself is in a state of disordered dissipation, as long as it can be described by a structured data model and probe grid, it reduces the joint entropy of the system in information theory by introducing mutual information $I(X; Y)$, thereby establishing a negative entropy order in the logical

space. Therefore, intelligence locks the conservation flow in the energy and matter dimensions and suppresses noise in the information dimension. It is the most magnificent rebellion in the physical universe.

Holographic Anti-Entropy theory issues a cold scientific judgment: the essence of intelligence is by no means prior-free probabilistic auto-regression. Rather, it is a reverse-work program spontaneously collapsed by a self-adaptive system under the gravitational dissipation of the universe sliding towards heat death, in order to minimize variational free energy. It is a "holographic negative entropy pump" that consumes its own high-frequency computing power and imports double negative entropy streams from the outside.

This paper aims to standardize this twenty-two-year scientific achievement mathematically and establish it as the physical constitution for systems science to govern and reshape physical reality in the digital intelligence era. To improve the rigor of this framework, we define the core terms formally before the main derivation:

Definition 1.1 (Holographic Anti-Entropy (HAE)). *Defined as the continuous reverse-work control process of a self-adaptive, self-organizing system far from equilibrium in a high-dimensional tensor phase space, which imports external negative entropy streams and performs internal topological renormalization to minimize Variational Free Energy against thermodynamic dissipation.*

Definition 1.2 (Digital Double Helix Ontology). *Defined as the product space $\langle D, A \rangle$ that physically carries and projects the governing operator Π in the silicon computing space. Here, D is a five-dimensional orthogonal spatiotemporal container (responsible for qualitative legislation and boundary constraints), and A is a discrete-time evolutionary dynamics algorithm (responsible for quantitative resolution and state updates).*

Definition 1.3 (Carbon-Silicon Topological Symbiosis). *Defined as the distributed double-loop feedback mechanism $\Phi - \Pi$ between the high-dimensional carbon-based metacognitive operator Φ and the low-dimensional silicon-based state convergence operator Π when governing open complex giant systems. Silicon executes rapid quantitative convergence within the axiomatic boundary, while carbon reconstructs the system axioms through metacognition outside the boundary to achieve self-evolution and leaps.*

Definition 1.4 (Conscience-Driven Mind OS). *Defined as the L0-level embodied cognitive system of the General Designer. It comprises a coupling loop of five crystalline dimensions: conscience (physiological pain residual attractor), high sensitivity (physical signal probe), sentient depth (neuronal resonance amplifier), fluid intelligence (experience-free algebraic abstractor), and metacognition (formal system amendment operator).*

2 Related Work and Theoretical Positioning

The Holographic Anti-Entropy (HAE) theory proposed in this study lies at the intersection of systems science, cybernetics, industrial engineering, and computational physics. To establish the theoretical position of this framework, we conduct a systematic dialogue with four major academic lineages:

1. Integration with the Tsien Hsue-shen School of Systems Science

The "metasynthesis method from qualitative to quantitative" proposed by Qian Xuesen in his later years is the core methodology of Hall of Metasynthesis (Metasynthesis), pointing out the "General Design Department" and the "Hall" as organizational entities for controlling complex giant systems. However, in specific engineering implementations, due to the lack of connection

between intermediate algebraic operators and micro-calculable state machines, this thought is often misunderstood as a qualitative "expert symposium."

HAE theory is an important advancement of the Tsien school: by constructing the digital double helix ontology $\langle D, A \rangle$, we compile qualitative experience into the physical extreme constraints of container D and delegate quantitative calculation to the bare-metal meta-operators of algorithm A . This runs the closed loop of "silicon-based rapid computation compensating for carbon-based computational deficits, guided by carbon-based metacognition" for the first time in engineering, making metasyntesis an executable, calculable, and objective physical system.

2. Dialogue with Complexity Theory and Computational Physics

In statistical physics and adaptive control, Karl Friston's "Free Energy Principle (FEP)" describes the survival of life as a process of minimizing variational free energy through active inference. Stephen Wolfram, through "Computational Irreducibility," points out that the future states of non-linear systems can only be deduced through step-by-step simulation.

HAE performs substantial theoretical expansion at the intersection of the two: FEP clarifies "why we resist entropy" (purpose), while HAE constructs "how to compute and control" (means) under the extreme non-homogeneous constraints of 100-billion-level discrete manufacturing. Through "physical gravity pruning" and "bitmap vectorization," we compensate for the explosion of computational irreducibility at the hardware level, translating the variational free energy optimization of active inference into prefix-sums and bit operations in bare-metal memory. This proves the calculable path of the Free Energy Principle in the governance of ultra-large-scale commercial/industrial entities.

3. Comparison with Industrial Engineering and Manufacturing System Theories

In classical industrial engineering, Wallace Hopp and Mark Spearman's "Factory Physics" used queuing theory and Markov chains to provide pioneering semi-physical descriptions of steady-state, localized manufacturing systems; Taiichi Ohno's Toyota Production System (TPS/JIT) achieved zero-inventory coordination at the physical level through kanban management.

However, the limitation of "Factory Physics" is that its underlying "stationary distribution and mean assumptions" inevitably collapse in the face of multi-level BOM entanglement and black swan fluctuations. The limitation of TPS is that it relies entirely on the propagation delay of physical kanban, facing information disconnection in the large-scale space of global value chains. HAE revolutionizes both: we deconstruct discrete manufacturing as a "spatiotemporal topological measure" flow field exhibiting non-equilibrium properties, utilizing ultra-real-time deduction algorithms to execute algebraic pruning before physical variations arrive. Instead of static flattening, we use high-frequency silicon computation to absorb external emergence in the logical space, achieving "superconducting" operation of cross-border value chains.

4. Inheritance and Transcendence of Classical Cybernetics

This study inherits Norbert Wiener's feedback control theory, W. Ross Ashby's Law of Requisite Variety, and Rudolf E. Kalman's observability conditions.

Ashby's law states that the complexity of the controller must not be lower than the complexity of the system being controlled. This locks the $O(N!)$ state space explosion and the Miller limit of the carbon-based brain in a deadlock. The HAE scheme natively solves this chasm: it uses the "single-brain singularity" to force logical complexity to collapse to $O(1)$ at the qualitative level, while executing bare-metal data-oriented designs inside the silicon system

to compensate for the quantitative calculations of $O(N!)$. Kalman observability requires the observability matrix to be full rank. HAE accordingly proposes the "reverse construction law," strictly limiting control power to the rigid observability of the physical bottom layer, dismantling the "digital nihilism" that relies purely on top-level dashboard metrics for governance.

3 Constitution: Axiomatic Deduction and Formal Proof of Tsien's "Open Complex Giant System" Theory

Qian Xuesen called for the creation of "systems science" and pointed out the "metasynthesis method from qualitative to quantitative" as the only channel to conquer complexity. However, due to the lack of algebraic operators and micro-calculable state machines, systems science has stayed in the passive paradigm of "qualitative description and post-hoc explanation" for half a century.

The fundamental innovation of Holographic Anti-Entropy theory lies in systematizing systems science itself for the first time, reconstructing it as a runnable, self-evolving "structural engine." Using "nodes (Node), topological relations (Topology), and transitions (Transition)" as the core algebraic basis, it connects the physical control loop from "teleology" to "evolutionary theory," completing the paradigm shift of systems science from "descriptive philosophy" to "calculable science." Starting from four basic axioms, this paper provides the first formal proofs of the four core pillars of Tsien's theory: metasynthesis, human-machine collaboration, the General Design Department, and the Hall of Metasynthesis.

Axiomatic Foundation

Axiom 3.1 (Methodology). *The only physical carrier for governing open complex giant systems is the digital double helix ontology $\langle D, A \rangle$.*

Axiom 3.2 (Ontology). *The carbon-based brain's bandwidth for concurrent information processing is locked by Miller's limit, necessitating the introduction of the silicon-based operator Π to compensate for factorial complexity.*

Axiom 3.3 (Evolution). *Based on Gödel's Incompleteness Theorems, any logically self-consistent formal system Π is destined to be unable to cover unknown variations, requiring the carbon-based high-order evolutionary operator Φ as a self-healing center outside the boundary.*

Axiom 3.4 (Capability). *According to Arrow's Impossibility Theorem, under multi-dimensional non-convex hard constraints, the group decisions of multiple equal subjects cannot aggregate into a global optimum.*

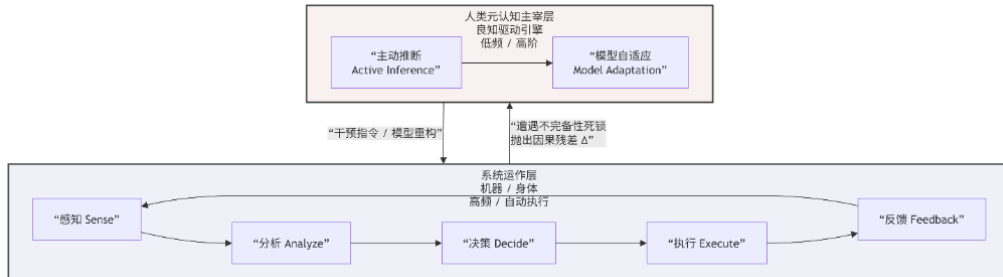


Figure 1: Carbon-Silicon Topological Symbiosis Architecture

Theorem 3.1: The Physical Entity of Metasynthesis is the Digital Double Helix Ontology $\langle D, A \rangle$

Proof. According to Axiom 3.2, the carbon-based brain cannot process the $O(N!)$ state space, so the silicon operator Π must be introduced to compensate for the compute limit. According to Axiom 3.1, the engineering entity of the operator Π is the digital double helix ontology $\langle D, A \rangle$ deeply coupled by the five-dimensional orthogonal spatiotemporal container D and the discrete-time step-by-step deduction algorithm A .

Here, the spatiotemporal container D is defined as the following five-dimensional orthogonal information manifold:

1. **Node:** Locates and decouples underlying physical, contractual, and value entities (solving "who the subjects and objects are").
2. **Topology:** Defines the Directed Acyclic Graph (DAG) and cross-dimensional mapping matrix of material, contract, and value flows (solving "how the logical structures connect").
3. **Constraints:** Restricts the phase space degrees of freedom via hard physical boundaries and flexible contractual inequalities (solving "where the physical extreme boundaries are").
4. **State Transition:** Discrete state changes triggered instantaneously by physical events, executing 1-to-3 orthogonal projections (solving "how the dynamic system changes").
5. **State Vector:** Instantaneous dynamical snapshots of supply potential, demand tension, and network rigidity (solving "where the system currently is").

This five-dimensional orthogonal container D satisfies strict MECE (mutually exclusive, collectively exhaustive) holographic completeness:

- **Mutually Exclusive:** Node, Topology, Constraints, State Transition, and State Vector describe system nodes, static connections, solution boundaries, discrete dynamic changes, and instantaneous state coordinates, respectively. Their algebraic degrees of freedom are mutually exclusive and orthogonal.
- **Collectively Exhaustive:** The state-space modeling of any open adaptive system is completely covered by its components (Node and Topology), feasible domain (Constraints), and evolutionary trajectory (Transition and State Vector). No holographic projection, from astronomical scales to organic networks and industrial value chains, can escape this five-dimensional representation.

The closed-loop operation of $\langle D, A \rangle$ fully realizes "metasynthesis from qualitative to quantitative" in engineering. $\square$

Theorem 3.2: Human-Machine Collaboration is a Double Necessity of Carbon-Based Computational Compensation and Boundary Breakthrough

Proof. The necessity of human-machine collaboration is proved by two independent logical paths:

- **Path A (Necessity of Computational Compensation):** According to Axiom 3.2 (Ontology), the carbon-based brain's concurrency bandwidth is locked by Miller's limit, placing it in a permanent state of computational deficit. Facing the $O(N!)$ state space explosion of open complex giant systems, the carbon-based brain cannot execute global optimization and real-time deduction. Therefore, the silicon operator Π must be introduced as an extension of the carbon brain. Silicon executes high-concurrency, low-latency

quantitative convergence within established boundaries; carbon is freed from calculation drudgery to return to its irreplaceable role—defining constitutional rules and boundaries. This is computational-level collaboration.

- **Path B (Necessity of Boundary Breakthrough):** According to Axiom 3.3 (Evolution), any logically self-consistent silicon double-helix system $\Pi = \langle D, A \rangle$ is limited by Gödel’s incompleteness, meaning logical dead ends always exist that cannot be self-proved. When the system hits this boundary, the silicon operator locks due to the lack of meta-rules for handling unknown variations, leading to spontaneous entropy increase. At this point, the carbon evolutionary operator Φ must be introduced, using its pain perception to capture the residual $\Delta(t)$, execute physical root-seeking, and remodel unknown variations into new axioms. Silicon closes the loop inside the boundary, while carbon heals and evolves outside the boundary. This is evolutionary-level collaboration.

Conclusion: Human-machine collaboration is the double necessity of combining carbon and silicon at the compute level (daily work under known rules) and the evolutionary level (self-healing breakthroughs on unknown boundaries). Its engineering entity is the carbon-silicon topological symbiosis $\Phi - \Pi$. This architecture realizes the necessity of silicon compensation in Qian’s “human-machine integration,” and the irreplaceability of carbon constitutional amendments at the evolutionary level. $\square$

Theorem 3.3: The General Design Department is the Inevitable Organizational Form to Achieve the “Single-Brain Singularity”

Proof. According to Axiom 3.1, the double helix ontology $\langle D, A \rangle$ governing complex giant systems requires deep integration of business rules, data topology, and solvers. The deep coupling of these three in the cognitive domain requires isomorphism in their physical carriers, meaning that business deconstruction, mathematical modeling, and system architecture must be completed simultaneously in one physical brain (single-brain singularity). According to Axiom 3.4, any collaborative model that attempts to split this into assembly lines (such as business experts + algorithm experts + IT architects) will inevitably fail due to Shannon information decay and $O(M^2)$ communication friction. Therefore, an independent “General Design Department” must be established to place a General Designer with this three-in-one capability to govern the operator’s architecture. $\square$

Theorem 3.4: The Hall of Metasynthesis is the Engineering Product of Metasynthesis and Human-Machine Collaboration

Proof. From Theorem 3.1, the entity of metasynthesis is $\langle D, A \rangle$; from Theorem 3.2, the entity of human-machine collaboration is $\Phi - \Pi$. When the carbon General Designer Φ needs to intervene in system evolution, they inject new strategic constraints into the five-dimensional container defined by D through interaction with Π . A then executes ultra-real-time deduction in memory and presents the parallel multi-universe simulation results to Φ for decision-making. This high-frequency interaction environment of “carbon directing, silicon computing” realizes the “Hall of Metasynthesis from qualitative to quantitative” in engineering. $\square$

Unified Proof: Necessary and Sufficient Conditions for a System to Govern Open Complex Giant Systems

Proof. A system possesses the complete capability to govern open complex giant systems if and only if it adopts $\langle D, A \rangle$ for metasynthesis (Theorem 3.1), builds the $\Phi - \Pi$ architecture

for human-machine collaboration (Theorem 3.2), establishes the General Design Department to achieve the single-brain singularity (Theorem 3.3), and constructs the Hall of Metasynthesis with a silicon engine at its core (Theorem 3.4). Any system violating this architecture will inevitably fail in physics and mathematics due to violating the axioms supporting these theorems. This completes the formal proof of Qian’s systems science theory. $\square$

4 Territory: The Unified Fractal Isomorphism from ”Human-Governed Systems” to ”Inorganic and Organic” Communities

Holographic Anti-Entropy theory breaks down the barrier between the subjective human society and the objective physical world. It points out that for all self-adaptive self-organizing systems far from equilibrium, their evolutionary trajectories—which resist thermodynamic dissipation and minimize variational free energy—exhibit absolute scale invariance in their algebraic topological structures.

This study projects and converges the core evolutionary dynamics onto three major scales of giant systems:

1. **Inorganic Self-Organization Scale:** Interstellar dust clouds in phase space spontaneously undergo centripetal collapse, fragmentation, and multi-stage mass accretion under gravity. The minimization of global gravitational potential energy constitutes the blind emergent will at this scale.
2. **Organic Active Community Scale:** Community organic substrates and active biomass flow irreversibly through the food web topology. Under temperature and survival pressure, community flocking behavior emerges. The evolutionary selection pressure of nature constitutes the passive will at this scale.
3. **Social and Digital Industrial Scale:** The 100-billion-level discrete manufacturing network is projected as multi-dimensional parallel manifolds after decoupling R&D, marketing, and delivery. Because human systems cannot withstand the high heat dissipation of natural trial-and-error evolution, the General Designer must actively extract the top-level strategic intention vector and inject it into the governing operator.

In cross-scale topological renormalization, all effective anti-entropy work strictly follows the unified scalar product equation:

$$W = \mathbf{F} \cdot \mathbf{S} \cdot \cos \theta \quad (1)$$

where the angle θ is the mismatch angle between the subjective intervention vector and the objective optimal causal trajectory of the system. The governing operator converges the non-linear pressures generated by the random collisions of micro-nodes, eliminating internal friction and heat dissipation (Q) by making $\theta \rightarrow 0$ (so $\cos \theta \rightarrow 1$). This grants micro-nodes the absolute freedom to release kinetic energy and do work within their own paths.

5 Sword-Forging: Engineering Weapons to Conquer ”Computational Irreducibility” and ”Emergence” Explosion

Stephen Wolfram expounded the ”Computational Irreducibility” of non-linear systems: system evolution cannot skip steps, nor can it be fitted by continuous high-order functions; the only path is step-by-step execution. However, the discrete phase space behind each step of execution exhibits an exponential explosion, posing a computing chasm for traditional operations research.

HAE theory rejects mystical talk and directly forges a silicon sword under the extreme load of 100-billion-level industrial networks. The core engineering breakthrough lies in the deterministic mastery of the twin problems of "computational irreducibility" and "emergence."

5.1 Computational Irreducibility and Emergence: Both the Problem and the Answer

Computational irreducibility declares that any attempt to predict the future of a complex system using simple formulas is destined to fail—you cannot take shortcuts; you must run it step by step. Emergence, on the other hand, reveals that when each irreducible step is executed honestly, macroscopic order spontaneously arises, making the whole greater than the sum of its parts.

In engineering practice, these two physical walls point to the only correct path: do not try to predict emergence, but design rules so that emergence serves you; do not try to bypass calculation, but optimize the speed of calculation to its extreme limits.

The core algorithm architecture of this system abstracts complex system logic into a finite number of critical decision points on the main path. Even if the main path of deduction contains only 30 core steps, the conditional branches after each step of calculation will still cause the solution to exhibit an emergent complexity of 2^{30} .

Theorem 5.1. *In a complex system that is computationally irreducible and exhibits non-linear emergence, step-by-step simulation is mathematically necessary and unique.*

Proof. 1. **Axiom of No Shortcut:** Suppose there exists a closed-form analytical equation that can calculate the future phase space coordinates without running each step of the system's state machines. If this equation exists, it means the system's dynamics are computationally reducible, directly violating the premise that the system is computationally irreducible. Thus, analytical shortcuts do not exist in physics.

2. **Church-Turing Thesis:** The emergence process of discrete phase spaces is equivalent to Turing machine execution. According to the Church-Turing thesis, a Turing-complete minimum operator set must include data processing (computation) and conditional transfer (**if-then-else** branch judgment). In discrete computing phase spaces, the only algebraic representation of non-linear phase transitions and solution boundary cuts is conditional transfer based on state causality.

Therefore, the step-by-step deduction structure of "sequential computation + conditional branch (**if-then-else**) + recalculation" is the only valid and necessary mathematical representation for solving computationally irreducible evolutionary processes. $\square$

This "sequential execution under non-linear interaction" ensures that the execution path of the program is the natural trajectory of the physical system evolving towards the optimal solution in the phase space. The system does not pursue a one-step absolute optimal solution, but honestly executes the "computation + IF-THEN-ELSE + recalculation" step-by-step deduction. The logically self-consistent and physically feasible result that finally emerges is the unique, deterministic optimal solution under all current constraints.

5.2 Physical Constraint Algebraic Pruning

We weld physical gravity constraints—such as time unidirectionality, mass-energy conservation, and system physical limits—directly into the bottom layer of the D container. By executing algebraic truncation at the very early stages of deduction, we freeze and filter out the vast majority of paths that have zero physical feasibility. This reduces the NP-hard phase space to a manageable size, achieving rapid convergence.

5.3 De-semanticized Data-Oriented Design

We completely abandon the traditional high-overhead, high-latency object-oriented memory model. Instead, we design de-semanticized, one-dimensional continuous compact arrays and binary bitmap vectorization calculations at the computational bottom. It deconstructs and projects complex system logic directly into binary 0s and 1s. Utilizing superconducting in-memory computing, we eliminate the transmission bandwidth deadlocks of the von Neumann architecture, enabling lock-free concurrent computation at the CPU cache level.

5.4 Experimental Setup and Baseline Comparison

To verify the performance of HAE theory and the newly reconstructed IPC DOD engine in discrete manufacturing networks under extreme loads, we build upon the author’s previous experience in implementing the first and second generation Intelligent Production Scheduling (IPS) systems at Lenovo’s proprietary factories and LCFC. We execute generational reconstruction and complete redevelopment to forge a generalized IPC engine, conducting crucial experiments on real-world data from LCFC (annual revenue exceeding 100 billion RMB, with more than 80

1. Data Source and Anonymization

The experimental samples use the actual operating data of LCFC. Due to core commercial confidentiality, the data was anonymized and de-identified during export (material codes converted to integer indexes, production line capacity to relative scalars, and order due dates to discrete time offsets), while preserving all original topological links and non-convex hard physical constraints. The data scale includes:

- Discrete daily demand orders: 500,000;
- Bill of Materials (BOM) levels: up to 20 levels, with total network nodes exceeding 2,000,000;
- Physical constraint nodes for equipment, lines, and molds: 150,000 inequality constraints.

2. Hardware Computing Environment

All calculations ran on a single, ordinary workstation (without using clusters or cloud computing):

- Processor: Intel Xeon 16-Core 3.2GHz;
- Memory: 128GB DDR4 RAM (the resident memory for IPC DOD bare-metal operation is 12GB);
- Operating System: Linux Centos 7.9.

3. Baseline Comparison and Physical Performance Reconciliation

We compared the IPC engine with the prevailing commercial scheduling systems (APS/ERP engines based on traditional relational databases and OOP heap memory architectures) of major international vendors under identical data conditions. The comparison metrics are shown in Table 1:

Table 1: Physical performance comparison between the IPC engine and traditional systems of major international vendors

Performance Metric	Traditional System (ERP/APS) ¹	IPC Engine (DOD Bare-Metal)
500,000 orders, 20-level BOM, 2M+ materials capacity netting & scheduling	Calculation crashes (out-of-memory) or takes several days	5 minutes (296 seconds) for global convergence
50,000 order core MRP calculation	3–8 hours	100.85 milliseconds
Underlying architecture	Table-driven, relational rows, OOP heap jumps	Data-Oriented Design (DOD), compact 1D arrays, SIMD parallelization
Cache hit rate & jump overhead	High Cache Miss (latency black hole); pointer jumps waste >85% of CPU cycles	100% Cache Hit ; no branch prediction failure pipeline flushes
Process auditability	Black-box statistical reports; logical paths cannot be traced	100% physical tracing ; causal reconciliation and DuckDB sandboxing

¹ The "traditional system of major international vendors" refers to the core industrial engines of global ERP/APS providers (such as SAP IBP, Oracle ASCP, and Blue Yonder ESP) that dominate the discrete manufacturing market. This represents the technical paradigm of current manufacturing enterprises. Their failure or extreme delays under our experimental dataset are not directed at a specific product, but reveal the inevitable collapse of the old paradigm when facing the $O(N!)$ state space explosion. This old architecture shares its theoretical roots with the "stationary distribution and mean assumptions" of *Factory Physics*, which assumes steady states and I.I.D. data, making its failure mathematically inevitable under Non-IID combinatorial explosions. The architectural features of these traditional systems (relational databases, OOP heap memory, nightly batch runs) are publicly documented industry standards. This conclusion is based on the declared technical paradigms of these systems and well-established computing limits, rather than proprietary code access.

4. Replicability and Open Algorithm Declaration

To ensure academic and industrial replicability, the pseudo-code for the core algebraic operators of the engine (including the branch-free prefetching algorithm, the SIMD prefix sum, and the bidirectional deduction algorithm) is provided in the Appendix. The anonymized simulation datasets and solver frameworks are open-sourced on GitHub and Gitee to support verifiable systems science research.

5.5 Crucial Experiment of Physical Materialized Closed-Loop Work

Under the extreme discrete gravity of LCFC, with annual revenues exceeding 100 billion RMB and 80

At the deduction endpoint, the system writes back the optimal self-healing solution rigidly, locking the physical state allocation. This is not a "recommendation report"; it is a digital order that the physical system must obey under the gravity of physical constraints. Using silicon-based computing, we outran variations on the time axis, compensating for the internal dissipation of human static perception through high-frequency information calculation.

6 Commander-Selection: The L0-Level Cognitive OS of the General Designer Governing Open Complex Giant Systems

The formal proofs and engineering execution of the eight physical laws reveal the most sacred carbon-based core of systems science: how does the cognitive model of the General Designer—the singularity of metasynthesis—collapse and emerge in the evolutionary path of life?

Systems physics issues a cold judgment: the General Designer is by no means an elite trained and selected by traditional bureaucratized education systems, because bureaucratized systems follow Conway's Law and local Nash game arbitrage. The General Designer is an anomaly that grows "wildly" under extreme pressure, driven by a raw feeling that even their own brain cannot control.

This L0 cognitive operating system comprises five crystalline dimensions that are highly "mutually exclusive" in the eyes of the public:

- **Conscience:** Sets the direction of convergence. Conscience is not a metaphysical moral label, but a physical gravity attractor embodied in the General Designer, manifesting as a physiological pain response to system dissipation, friction, and the suffering of peers, setting an inviolable global objective function for all computation.
- **High Sensitivity:** The frontline physical probe. Intensely aware of the physical world and biological nodes, it instantly captures weak signals obscured by environmental noise. This is the starting point of all cognition and pain, and the physical probe to capture the holographic residual $\Delta(t)$ between reality and prediction.
- **Sentient Depth:** The ultimate negative entropy fuel. Humans are essentially predictive models. Sentient depth decodes the signals received by high sensitivity—the struggles of micro-nodes generate resonance in the mirror neuron system, driving fluid intelligence to perform extreme search in the phase space.
- **Fluid Intelligence:** The optimization engine in phase space. In completely unfamiliar territory, they do not rely on past experience but use pure reasoning and pattern recognition. Facing unprecedented complex networks, they strip away semantic surface details to abstract the physical essence and perform step-by-step deduction.

- **Metacognition:** The self-referential monitoring and "constitutional amendment" operator. It constantly audits the residual between the model and reality. At the singularity where the formal system locks up, it breaks the old framework and executes topological reorganization of the axiomatic system.

These five traits rarely converge in a single individual. They are the rule-breakers who reverse Conway's Law and bureaucratic structures.

The ultimate altruistic value of HAE's formal proofs and open-source code is to cultivate the soil and protect these fragile singular sparks of the new generation: we use a cold, deterministic silicon foundation to compensate for the pain of predecessors who bridged system gaps with their flesh. This allows high-sensitivity, high-perception thinkers to bypass bloody neuronal remodeling and directly wield the "eight digital scalpels" to govern broader cosmic complexity on superconducting self-healing tracks.

7 Holographic Anti-Entropy and the Eight Unified Laws of Systems Science

The following eight laws declare an irreversible formal judgment of systems science in the highest phase space of topological algebra, measure theory, and adaptive control, proving the inevitability of Qian's OCGS theory in the digital intelligence era.

The First Law (Teleology): The Unified Law of Anti-Entropy Work

The self-organizing survival of open complex giant systems is essentially the process of the governing operator injecting structured negative entropy into a high-dimensional tensor phase space to offset the system's internal thermodynamic dissipation. Its dynamic evolution and cognitive loop are jointly constrained by three physical axioms:

$$dS = d_i S + d_e S \quad (d_e S < 0, |d_e S| > d_i S \implies dS < 0) \quad (2)$$

$$S_{\text{negative}} = -S = -k_B H(X) = k_B \sum_{i=1}^n P(x_i) \log_2 P(x_i) \quad (3)$$

$$F = D_{\text{KL}}(q(s) \parallel p(s|o)) - \ln p(o) \geq -\ln p(o) \quad (4)$$

Formal Proof that Governance is a Precondition for Observation and Simulation

In non-linear giant system dynamics, the system must establish governance (setting rigid algebraic constraints and coordinates) as a precondition to close the causal loop of "observation, simulation, and governance." If the rigid algebraic constraints of the governing operator Π are missing, system observation and simulation will inevitably distort. The mathematical proof is as follows:

1. **Identifiability Deadlock of Closed-Loop Feedback:** Let the controlled system be Ω . The state transition equation of its micro-nodes is:

$$y(t) = Gu(t) + w(t) \quad (5)$$

where $u(t) \in \mathbb{R}^d$ is the actual action (control input) of micro-nodes, $w(t)$ represents the unobservable disturbances (internal noise), and $G \in \mathbb{R}^{m \times d}$ is the system causal transfer operator to be observed and identified. In the absence of a governing operator to

impose orthogonal algebraic constraints (such as "human governance" or "disordered self-governance"), the actual input $u(t)$ of each node is not an independent variable, but a passive feedback response of local agents to the noise and state of the system:

$$u(t) = Ky(t) + \epsilon(t) \quad (6)$$

Combining the two equations to eliminate the output $y(t)$, we obtain the actual input:

$$u(t) = (I - KG)^{-1}(Kw(t) + \epsilon(t)) \quad (7)$$

At this point, the control input $u(t)$ is highly correlated with the internal noise $w(t)$, resulting in a strong collinear coupling:

$$\text{Cov}(u, w) \neq 0 \quad (8)$$

2. **Causal Distortion of Passive Observation:** When the observation system attempts to identify the system causal model $\hat{G}$ through passive data collection of $\{u(t), y(t)\}$, the parameter estimator is:

$$\hat{G} = \left(\sum_{t=1}^T y(t)u(t)^T \right) \left(\sum_{t=1}^T u(t)u(t)^T \right)^{-1} = G + \left(\sum_{t=1}^T w(t)u(t)^T \right) \left(\sum_{t=1}^T u(t)u(t)^T \right)^{-1} \quad (9)$$

Even if the sample size $T \rightarrow \infty$, the estimation bias term cannot be zero:

$$\lim_{T \rightarrow \infty} \hat{G} = G + R_{wu}R_{uu}^{-1} \neq G \quad (10)$$

This produces a local collinearity identification deadlock. Passive observation inevitably misinterprets the correlation of noise as causality, yielding a distorted transfer function $\hat{G}$. The simulation deduction based on this model will diverge exponentially within the maximum Lyapunov Horizon of the system's forward run, leading to simulation illusions:

$$\tau_{\max} = \frac{1}{\lambda_{\max}} \ln \left(\frac{\Delta_{\max}}{\delta_0} \right) \rightarrow 0 \quad (\text{when } \lambda_{\max} \gg 0) \quad (11)$$

Under un-governed states, the maximum positive Lyapunov exponent $\lambda_{\max}$ explodes factorially, collapsing the maximum simulatable horizon $\tau_{\max}$ to zero. The observation space distorts, and the governance work vectors cancel each other orthogonally:

$$Q = \sum_{i=1}^n \|V_i\| - \left\| \sum_{i=1}^n V_i \right\| \geq 0 \quad \text{and} \quad W = \mathbf{V}_m \cdot \mathbf{V}_\pi \cdot \cos \theta \rightarrow 0 \quad (\theta \rightarrow 90^\circ) \quad (12)$$

3. **Orthogonal Decoupling of Governance (Identifiability Condition):** The only solution for identifiability is to introduce the governing operator Π to impose a set of orthogonal physical constraints independent of system noise (such as rigid strategic allocations $u_\Pi(t)$ defined by the General Designer):

$$u(t) = u_\Pi(t) + u_{\text{local}}(t) \quad \text{where } \text{Cov}(u_\Pi, w) = 0 \quad (13)$$

The governing operator cuts the feedback dependency of the micro-inputs on internal disturbances through rigid algebraic constraints, making the input self-covariance matrix R_{uu} full rank. This decouples inputs from noise, allowing the estimator to converge to its true physical value:

$$\hat{G} \rightarrow G \quad (14)$$

In terms of information theory, the mutual information $I(G; O)$ between the observation system and the real system satisfies:

- Under un-governed self-governance: $I(G; O) \rightarrow 0$ (observation degenerates into noise projection, failing to extract real causality);
- Under governed constraints: $I(G; O) > 0$ (physically identifiable).

Therefore, we must first establish the rigid boundaries of the governing operator Π to suppress the maximum Lyapunov exponent to the stable domain ($\lambda_{\max} \leq 0$) and decouple feedback noise. Only then can we validate holographic observation and dynamic simulation. **Governance is the absolute precondition for observation and simulation to possess physical validity.**

The Second Law (Ontology): Ashby's Limit and Silicon-Based Compensation

The number of state variations of the controlled system explodes factorially with the number of nodes, while the concurrent information processing bandwidth of the carbon-based brain is locked by Miller's limit, placing it in a permanent computational deficit. The collapse of system control and machine compensation are governed by four irreducible physical equations:

$$V_c \geq V_s \approx O(N!) \quad (15)$$

$$C_{\text{carbon}} = 7 \pm 2 \ll V_s \quad (16)$$

$$\text{Non-IIDness} = f(\text{Coupling}(\text{Intra}, \text{Inter}), \text{Heterogeneity}(\text{Dist}, \text{Temp-Spat})) \quad (17)$$

$$\max_{p_i} H(X) = - \sum_{i=1}^n p_i \ln p_i \quad \text{s.t.} \quad \sum p_i = 1, \quad \sum p_i g(x_i) = \bar{G} \quad (18)$$

Due to the non-independent and identically distributed (Non-IIDness) nature of entities and the spontaneous maximum entropy degradation under high entropy states, the state space of the system expands rapidly to $O(N!)$. Under the gravity of $C_{\text{carbon}} \ll V_s$, we must enforce physical extreme constraints and algebraic truncation at the bottom layer of the D container, transferring control power rigidly to the silicon operator Π to compensate for the limits of carbon.

The Third Law (Methodology): Dimensional Reduction Isomorphism of the Digital Double Helix Ontology

The physical carrier of the governing operator in silicon memory is the digital double helix ontology $\langle D, A \rangle$ composed of the five-dimensional orthogonal container D and the discrete-time step-by-step deduction algorithm A . In cybernetics and information processing, the system obeys:

$$\Pi = \langle D, A \rangle : \Omega_{O(N!)} \longrightarrow \Omega_{O(k)} \quad (19)$$

$$\text{rank}(\mathcal{C}) = \text{rank} [B, AB, A^2B, \dots, A^{n-1}B] = n \quad (20)$$

$$h : \Omega \longrightarrow D \quad (\text{Conant-Ashby Homomorphic Mapping}) \quad (21)$$

$$f_{\text{compute}} > f_{\text{perturbation}} \quad (\text{Nyquist Sampling Inequality}) \quad (22)$$

The computational frequency of the controller f_{compute} must outrun the external disturbance frequency $f_{\text{perturbation}}$. Through the scale-invariant fractal property ($P(k) \propto k^{-\gamma}$) and controllability full-rank constraints, we utilize a unified $\langle D, A \rangle$ topological grammar across different layers to absorb the $O(N!)$ state space explosion in memory.

The Fourth Law (Capability): Causal Logic Convergence of the Single-Brain Singularity

Under multi-dimensional, non-convex hard constraints, the group decisions of multiple independent administrative subjects cannot aggregate into a global optimum according to Arrow's Impossibility Theorem. The core logic of conquering complexity requires forcing a reverse Conway maneuver during the design phase, collapsing organizational will into a "single-brain singularity" that integrates business, data, and architecture:

$$\max F(x)_{\text{global}} \neq \sum \max f_i(x)_{\text{local}} \quad \text{and} \quad U_{\text{global}} \neq \sum U_{\text{local}} \quad (23)$$

$$\text{Complexity}_{\text{comm}} = O(K^2) \gg \text{Complexity}_{\text{singularity}} = O(1) \quad (24)$$

If K nodes are introduced for networked decision-making, the communication complexity will explode quadratically ($O(K^2)$). To eliminate channel dissipation, we must force the global logic to converge into a single-brain singularity, collapsing the decision complexity to $O(1)$ and establishing an orthogonal data base through absolute "logical authority."

The Fifth Law (Mechanism): Algebraic Topological Isomorphic Mapping of Organizations

The collaboration mechanism of system digital transformation must be topologically isomorphic and nested hierarchically with the final silicon-based digital brain. In terms of macroscopic emergence and microscopic allocation, the system obeys von Bertalanffy's wholeness equation, Haken's synergetics slaving principle, and spectral decomposition convergence:

$$V_{\Omega} = f(R) \cdot \sum_{i=1}^K V_i \neq \sum_{i=1}^K V_i \quad (25)$$

$$V_{\Omega} = M \cdot \Pi [S_1 \otimes S_2 \otimes \cdots \otimes S_n] \quad (26)$$

$$P = U \Sigma V^T \rightarrow P_k = U_k \Sigma_k V_k^T \quad (\text{SVD Spectral Truncation}) \quad (27)$$

The system reorganizes the global collaboration pressure upward through the governing operator Π in memory, eliminating the cross-domain coordination burden of micro-nodes S_i . By issuing renormalized micro-objectives and algorithms (m_i^*, π_i^*), it grants micro-nodes the absolute freedom to focus on their own specialized paths within defined orthogonal allotments.

The Sixth Law (Pathology): The Wiener-Kalman Observability Boundary Law (Reverse Construction Law)

The dimension of the control domain is strictly limited by the dimension of the micro-observation space. The construction path must follow a one-way, irreversible anchoring from the bottom physical reality to the high-dimensional formal space, satisfying the observability condition and Rosen's modeling relation:

$$\dim \mathcal{C} \leq \dim \mathcal{O} \quad (28)$$

$$\text{Encoding} \circ \text{Causal}_{\text{NS}} = \text{Implication}_{\text{FS}} \circ \text{Encoding} \quad (29)$$

If the planned solution calculated by the formal system (FS) cannot be written back rigidly (Decoding / Write-Back) to the natural system (NS) through Allotment to lock the physical allocation, the logic will remain suspended at the physical layer. The construction path must enforce "reverse physical construction"—first opening up the encoding and write-back of execution-layer probes, before constructing the macroscopic control tower.

The Seventh Law (Dynamics): Alignment of Executive Power and Objective Logic Fields

The system acceleration and actual work of transformation are jointly determined by the administrative force and algorithm gravity, constrained by convex optimization boundaries and the vector scalar product theorem:

$$m \cdot \mathbf{a} = \mathbf{F}_M + \mathbf{F}_\Pi - \mathbf{f} \quad \text{and} \quad W = \mathbf{V}_m \cdot \mathbf{V}_\pi = \|\mathbf{V}_m\| \|\mathbf{V}_\pi\| \cos \theta \quad (30)$$

$$\lambda_j = \frac{\partial V_\Omega}{\partial b_j} \quad (31)$$

The actual effective work done in system transformation depends on the angle θ between the power will vector $\mathbf{V}_m$ and the objective optimal evolution path $\mathbf{V}_\pi$ of the algorithm. By solving the shadow prices of the physical boundaries b_j (the Lagrange multipliers λ_j), the governing operator directs administrative power precisely to the system's bottleneck points, forcing $\theta \rightarrow 0$ (so $\cos \theta \rightarrow 1$) and achieving lossless work.

The Eighth Law (Evolution): Anti-Heat-Death Evolution of Self-Adaptive Systems

When the static silicon-based axiomatic system experiences logical deadlocks at its incompleteness boundary, a carbon-based metacognition equipped with embodied conscience pain must be introduced to execute non-autoregressive axiom rewriting, achieving the evolution and leap of the adaptive organism:

$$\frac{dS}{dt} = \frac{d_i S}{dt} + \frac{d_e S}{dt} \quad \left(\frac{d_e S}{dt} < 0, \left| \frac{d_e S}{dt} \right| > \frac{d_i S}{dt} \implies \frac{dS}{dt} < 0 \right) \quad (32)$$

$$\Pi_{t+1} = \Phi \left(\Pi_t, \sum \Delta_{\text{critical}} \right) \quad (33)$$

$$\Delta(t) = \|S(t) - S_{\text{pred}}(t)\| \quad (34)$$

$$\langle e^{-\beta W} \rangle = e^{-\beta \Delta F} \quad (\text{Jarzynski Non-Equilibrium Work Equality}) \quad (35)$$

When external critical macroscopic uncertainty residuals trigger a system deadlock, a carbon high-order evolutionary operator Φ (the subjective true perception intervention of the General Designer) must be introduced. The activation of Φ is not a random or subjective epiphany, but a physical phase transition process that the self-adaptive control system must undergo to combat heat death. We define three strictly observable triggering conditions for the activation of Φ :

1. **Non-linear Divergence of Prediction Residuals (Failure of Residual Convergence):** As the system runs, the prediction residual $\Delta(t)$ diverges non-linearly, rendering the prediction model invalid:

$$\lim_{t \rightarrow T_{\text{crit}}} \Delta(t) = \|S(t) - S_{\text{pred}}(t)\| \rightarrow \infty \quad (36)$$

2. **Non-Convergence of Strategy within Finite Time:** Within a given finite deduction time T , the system cannot achieve target convergence through any known algorithm $a \in \mathcal{A}$ corresponding to the current axiom set:

$$\forall a \in \mathcal{A}, \quad \|S(t+T) - S^*(t+T)\| > \epsilon \quad (37)$$

3. **Logical Deadlock of System Axioms (Empty Feasible Domain):** The formal axiom set $\mathcal{K}_t$ inside the controller generates an inference deadlock, causing the physical feasible domain in silicon memory to collapse directly into an empty set:

$$\mathcal{K}_t \vdash P \wedge \neg P \implies \mathcal{S}_{feas} = \emptyset \quad (38)$$

When these three conditions are met simultaneously, the variational free energy of the system will cross the critical phase transition barrier. Because the open adaptive system must maximize negative entropy import ($d_e S/dt < 0$) to combat heat death, at the singularity of logical self-consistency rupture, the system undergoes a topological phase transition driven by the carbon designer's "pain" (the conscience attractor) and the strategic push activation potential Λ .

The evolutionary operator Φ executes non-autoregressive axiom rewriting:

$$\Phi : \mathcal{K}_t \xrightarrow{\Lambda} \mathcal{K}_{t+1} \quad (39)$$

The new axiom system reopens the feasible solution manifold, eliminating the original constraint deadlock. The system converges once more and completes its generational leap.

This law establishes the ineliminability of carbon metacognition in the system: if one day an autonomous silicon system can independently complete the above generational axiom reorganization without any carbon intervention, then this law will be disproved. The creator of HAE believes the probability of this event is near zero—not because silicon is not powerful enough, but because the projection of Gödel's Incompleteness Theorems in the physical world implies that any closed formal system inevitably possesses a deadlock boundary that requires external intervention to cross.

8 Conclusion: Wisdom as the Instrument, Goodness as the Engine

The gravitational dissipation of the physical universe contains no mercy for human systems; isolated systems on the one-way track of time will eventually slide into the abyss of entropy increase and death. However, HAE's formal proofs and twenty-two years of clinical experiments demonstrate to the world:

Establishing global causal connections, practicing carbon-silicon topological symbiosis, anchoring the gravity of evolution with one's own true perception (conscience), and passing down the coding genes of exploring truth and fighting chaos from generation to generation—this is the greatest, most magnificent negative entropy energy we, as noble intelligent life, can deliver to this world against the vast, desolate background of a heat-death universe.

What remains is no longer a small workshop relying on the difficult struggles of individual heroes, but a digital organism that has eliminated all internal dissipation, can breathe freely, and can self-heal and evolve indefinitely. It works day and night, using the purest light of reason to cultivate an oasis of negative entropy in this dying cosmic desert.

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Appendix: Meta-Operators of the Holographic Anti-Entropy Core Algorithms

To ensure academic and industrial replicability, the pseudo-code for the core algebraic operators of the engine (including the branch-free prefetching algorithm, the SIMD prefix sum, and the bidirectional deduction algorithm) is provided in the Appendix:

Appendix A: Branch-Free Priority Netting and Temporal Consumption Algorithm

In the execution phase (IOP), to eliminate the CPU branch misprediction overhead caused by multi-level conditional branch sorting, the algorithm packs multi-dimensional order attributes into a single 64-bit unsigned integer for flat sorting. Meanwhile, in temporal deductions, algebraic operators are used to eliminate if-else conditional judgments, supporting CPU auto-vectorization:

Algorithm 1: Branch-Free Priority Netting

Input:

Orders set $O = \{o_1, o_2, \dots, o_m\}$

Supply time-series $S = [s_1, s_2, \dots, s_T]$

Output:

Updated Supply S , and Allocated quantities for each order.

1. For each order o_i in O :
 2. Pack multi-dimensional priorities into a single 64-bit unsigned integer:
 $P_i \leftarrow (\text{committed_bit} \ll 62) \mid (\text{tier_bits} \ll 60) \mid (\text{due_bits} \ll 44) \mid$
 $(\text{pri_bits} \ll 28) \mid (M_{\text{rev}} - \min(M_{\text{rev}}, \text{Revenue}_i))$
 3. Sort O in ascending order of P_i using standard comparison-free sorting or quicksort.
 4. For each sorted order o_i in O_{sorted} :
 5. $D_{\text{req}} \leftarrow o_i.\text{quantity}$
 6. $t_{\text{due}} \leftarrow o_i.\text{due_day}$
 7. For $t = t_{\text{due}}$ down to 0:
 8. // Eliminate conditional jump using algebraic min / CMOV:
 $q_{\text{alloc}} \leftarrow \text{std::min}(S[t], D_{\text{req}})$
 $S[t] \leftarrow S[t] - q_{\text{alloc}}$
 $D_{\text{req}} \leftarrow D_{\text{req}} - q_{\text{alloc}}$
 $o_i.\text{allocated}[t] \leftarrow q_{\text{alloc}}$
 9. If $D_{\text{req}} \leq 0$ Then Break
-

Appendix B: Parallel Prefix Scan & Segment Tree Associative Merger

For the same material node, the inventory level and Available-to-Promise (ATP) calculation are achieved through Blelloch parallel scan and segment tree merging. Under multi-site collaboration, the state of interval nodes is described by local demand deficit and local supply surplus vectors, solved in parallel within $O(\log T)$ using associative merging operators:

Algorithm 2: Parallel Prefix Scan & Segment Tree Associative Merger

Input:

Daily net supply/demand array $\text{Net}[T] = \text{Supply}[T] - \text{Demand}[T]$
DAG Sourcing topology matrix $\text{eta}[K][K]$ for K collaborative sites

Output:

OnHand time-series and global deficit/surplus vectors.

Part I: Parallel Blelloch Scan (Time Complexity: $O(\log T)$)

1. // Up-Sweep (Reduction Phase)
For $d = 0$ to $\log_2(T) - 1$:
Parallel For $i = 0$ to $T - 1$ step $2^{(d+1)}$:
 $\text{Net}[i + 2^{(d+1)} - 1] \leftarrow \text{Net}[i + 2^d - 1] + \text{Net}[i + 2^{(d+1)} - 1]$
2. Set $\text{Net}[T - 1] \leftarrow 0$
3. // Down-Sweep (Distribution Phase)
For $d = \log_2(T) - 1$ down to 0 :
Parallel For $i = 0$ to $T - 1$ step $2^{(d+1)}$:
 $t \leftarrow \text{Net}[i + 2^d - 1]$
 $\text{Net}[i + 2^d - 1] \leftarrow \text{Net}[i + 2^{(d+1)} - 1]$
 $\text{Net}[i + 2^{(d+1)} - 1] \leftarrow t + \text{Net}[i + 2^{(d+1)} - 1]$

Part II: Associative Merger Operator (oplus) for Segment Tree Nodes

Let $u_L = (\text{Deficit}_L, \text{Surplus}_L)$ and $u_R = (\text{Deficit}_R, \text{Surplus}_R)$ be child nodes.
The merged parent node $u_{\text{Parent}} = u_L \text{oplus } u_R$ is defined by:

4. Initialize $\text{temp_deficit} \leftarrow \text{Deficit}_R$, $\text{temp_surplus} \leftarrow \text{Surplus}_L$
 5. // Match deficits of R with surpluses of L along DAG paths (precedence order):
For each destination site i in $[0, K-1]$:
 For each source site j in $\text{Parents}(i)$:
 $q_{\text{trans}} \leftarrow \min(\text{temp_deficit}[i], \text{temp_surplus}[j] * \text{eta}[j][i])$
 $\text{temp_deficit}[i] \leftarrow \text{temp_deficit}[i] - q_{\text{trans}}$
 $\text{temp_surplus}[j] \leftarrow \text{temp_surplus}[j] - q_{\text{trans}} / \text{eta}[j][i]$
 6. $\text{Deficit_Parent}[i] \leftarrow \text{Deficit}_L[i] + \text{temp_deficit}[i]$
 7. $\text{Surplus_Parent}[j] \leftarrow \text{Surplus}_R[j] + \text{temp_surplus}[j]$
 8. Return $u_{\text{Parent}} = (\text{Deficit_Parent}, \text{Surplus_Parent})$
-

Appendix C: OTP Bidirectional Scheduling & Zero-Heap Rollback Stack

To eliminate the expensive memory allocation overhead during finite-capacity DFS path searching and preemption rollback, the algorithm records transaction states in a zero-heap pre-allocated thread-local stack, combining forward soft allocation (searching for the earliest delivery date) and backward rigid pull to achieve lock-free delivery confirmation and preemption:

Algorithm 3: OTP Bidirectional Scheduling & Zero-Heap Rollback Stack

Input:

Order demand: due_day , duration , wc_id
Flat 2D Spatiotemporal Capacity Grid: $\text{CapGrid}[\text{WC_Num} * \text{Horizon}]$
Thread-Local Rollback Stack: TLS_Stack

Output:

Planned delivery day or schedule rejection.

```

1. // Step 1: Forward Soft Allocation (Search for PSD / Best Can Do)
   PSD <- -1
   For t = request_day to Horizon - 1:
       offset <- wc_id * Horizon + t
       If CapGrid[offset] >= duration Then
           PSD <- t
           Break
   If PSD == -1 Then Return "Insufficient Capacity"

2. // Step 2: Backward Rigid Pull Scheduling & Preemption
   TLS_Stack.Clear()
   success <- False
   offset <- wc_id * Horizon + PSD
   If CapGrid[offset] >= duration Then
       // Deduct capacity and record action in TLS_Stack for potential rollback
       CapGrid[offset] <- CapGrid[offset] - duration
       TLS_Stack.Push(wc_id, PSD, duration, 0.0)
       success <- True
   Else
       // Run SWAP Preemption: search for lower-priority orders in PSD
       For each order o_low in CapGrid[offset].orders where priority(o_low) < priority(current_order)
           Rollback o_low capacity, push action to TLS_Stack
           Try rescheduling o_low to PSD + 1
           If o_low rescheduled successfully Then
               Allocate current_order to PSD
               success <- True
               Break

3. If success == True Then
   TLS_Stack.Commit() // O(1) clear, accept allocation
   Return PSD
Else
   TLS_Stack.Rollback(CapGrid) // O(1) restore to original state
   Return "Schedule Failed"

```
